

CS-544: Topics in Networks

Mid Semester Examination

February 2024

Total time- 120 minutes, Marks- 50

- ✓ 1. Explain the major benefits of using MPLS in today's IP networks? What is a Forwarding Equivalence Class (FEC)? Explain how MPLS can be used for traffic engineering purposes. (1+2+1 marks)
- ✓ 2. What is a work-conserving system? Do we need non-work conserving disciplines? Why or Why not? (1+2 marks)
- ✓ 3. What is internal blocking? Can a multistage crossbar suffer from internal blocking? Justify your answer. (1+2 marks)
4. Consider a first generation switch. The bus capacity is given as $\langle 133 \text{ MHz}, 64 \text{ bits} \rangle$. The main memory bus capacity is $\langle 1000 \text{ MHz}, 64 \text{ bits} \rangle$. You have a 3 GHz CPU with a requirement of 200 clock cycles per instruction to process a packet. Assume only one instruction. For a packet of size 500 bytes with 20 byte header, find the time taken to forward it from one IC link to another. Link rate for each IC link is 100 Mbps. Find the throughput and the number of line cards/ICs that can be supported by this system. (5 marks)
- ✓ 5. For a switch with buffers at the input queue, derive the theoretical switch utilisation. (6 marks)
- ✓ 6. Which queueing system would provide better performance: an M/M/3/300/100 system or an M/M/3/100/100 system? Justify your answer. (3 marks)
- ✓ 7. "For most scheduling disciplines, obtaining an Absolute Fairness Bound (AFB) is usually much harder than obtaining Relative Fairness Bound (RFB)". Justify. (2 marks)
- ✓ 8. How does connection admission control depend upon the scheduling algorithm used? (2 marks)
- ✓ 9. Consider a router's output queue (modeled as an M/M/1 queue) that can serve 2000 packets per second (with exponentially distributed packet length). If packets arrive based on a Poisson process with rate 1200 packets per second, determine the values of
 - i. Average total delay
 - ii. Average waiting time in the queue
 - iii. Total number of jobs in the system
 - iv. Total number of jobs in the queueing

$$\begin{array}{r} 125 \\ 1200 \\ \hline 3 \times 1000 \times 1000 \\ 5 \times 1000 \end{array} \quad \begin{array}{r} 5 \\ 4 \\ \hline \end{array}$$

— x P.T.O x —

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End Semester Examination

30 April 2024

Total time- 180 minutes, Marks- 80

Closed Book and Closed Notes; No Cell Phone; ONLY Scientific Calculator; No Sharing of Calculator;
No discussions with students enrolled in class.

1. State reasons why IP addressing is not geographic? (2 marks)

2. a. Given below are the specifications of a first generation router, find the total amount of time required to forward a packet from one IC to another. (4+2 marks)

The size of packets is 60 bytes with a 20 byte header. The capacity of the bus is <145MHz, 32 bits> and the main memory bus capacity is <1000MHz, 32 bits>. The router has a 4GHz CPU and to process a packet, 200 clock cycles are required per instruction. Assume only one instruction.

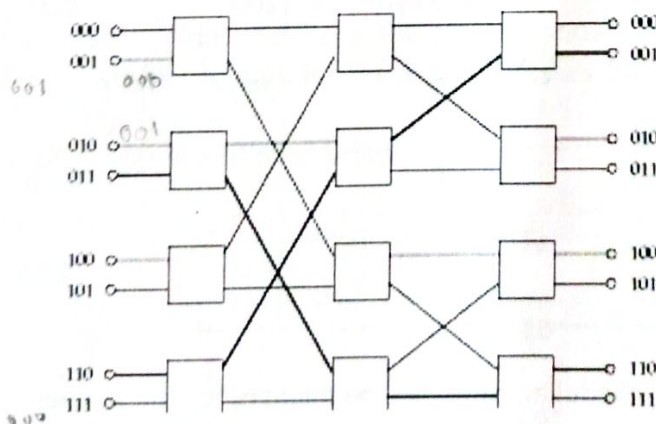
b. Using the above given data, if each IC has a requirement of 100 Mbps, calculate the number of ICs that can be supported in the given architecture.

3. Consider the following banyan switch:

a. Do packets destined for output ports 000 and 001 at input ports 001 and 010 collide?

b. Do packets destined for output ports 001 and 000 at input ports 001 and 111 of a banyan collide? (3+3 marks)

Show how you arrived to your answer.



4. The time between requests to a web server is measured and found to approximately

follow an exponential distribution with a mean time between requests of 8 ms. The time required by the server to process each request was also found to roughly follow an exponential distribution with an average service time of approximately 5 ms.

a. What is the percentage of time the server is working? (2x3 marks)

b. What is the average response time observed by the users making requests on this server?

c. If the requests are now made to two different servers with same average service time, how will the performance change?

5. A few wireless nodes are sending data to a central node (sink). The inter arrival time of the packets at the sink is exponentially distributed. Due to different packet sizes and structures, service times are exponentially distributed. If 125 packets arrive per second and the sink takes about 5 ms to process, find the following:

(2x5 marks)

- Sink utilization
- Mean number of packets in sink
- Mean time spent in sink
- Probability that there are 10 packets in the sink
- Probability of buffer overflow, if sink had 5 buffers

$$\rho = \lambda \cdot \mu + \dots + \lambda \cdot \mu^L$$

$$\lambda \cdot (1 - \rho^L)$$

6. Consider the following set of four connections (A, B, C, and D) that have packets arriving at specified instances, with specified packet length (in bytes). Assume that the link speed is 10 Mbps. All times are specified in microseconds.

Connection	T=0 microseconds	T=100 microseconds	T=400 microseconds
A	100 bytes	200	800
B	150	750	----
C	200	---	600
D	---	---	1200

Determine the scheduling order of the packets for the connections using Weighted Fair Queuing (with equal weights). Show all the details of calculation, including the rate of increase of round number and iteration deletion wherever applicable.

(10 marks)

8. Define the notion of vacations in an M/G/1 queue. Derive the Average Waiting Time (W) for an M/G/1 system with vacations, where arrival/departure parameters are as listed above and $E[V]$ and $E[V^2]$ denote the first and second moments of the vacation time. (2+8 marks)

9. Consider the slotted Aloha MAC protocol used for a channel shared by N nodes. Let λ denote the Poisson packet arrival rate per slot, to a node's transmitter queue. Let p_0 denote the retransmission probability of a backlogged node in a given slot. Assume that each node's transmit buffer can hold at most THREE packets, including the one for which transmission is being attempted. Use a Markov model to represent the node's behavior. Using this model, explain how we can determine the system throughput and average packet delay.

You can state any other simplifying assumptions as needed (e.g. the probability of more than one packet arriving in a slot is negligibly small.) (10 marks)

10. Consider an M/M/∞ queuing system with an infinite number of servers, with parameters λ and μ . In such a system, all arriving jobs being receive service immediately. The service rate is μ when there are n jobs in the system. Derive the expressions for average number of jobs in the system (N) and average total delay (T). Is the system always stable? (4+2 marks)

10. For the given table below, find the following: (4 marks)

Prefix Label	Prefix	
P ₁	0*	✓
P ₂	10*	✓
P ₃	111*	✓
P ₄	10001*	✓
P ₅	1*	✓
P ₆	1001*	✓
P ₇	101000*	✓
P ₈	1010000*	✓

Draw a fixed stride multibit tree using the prefixes shown in the Table. How many memory accesses will be required for the 8-bit addresses 10011000 and 10100011?

11. Consider the following classification table: (4 marks)

Rule	F ₁	F ₂
R ₁	0*	10*
R ₂	0*	01*

R ₃	0*	1*
R ₄	00*	1*
R ₅	00*	11*
R ₆	10*	1*
R ₇	11*	00*
R ₈	*	00*

For the rules given in the above table, construct a set pruning trie. What is the best matching rule for a packet with $F_1 = 0011$ and $F_2 = 0011$?

12. State the disadvantages of using a TCAM. Given an 8-bit wide TCAM, represent the rules specified in the Table in Question 11. (2+4 marks)

more transistors

less density (bits/area)

high electricity consumption